Prototype Submission Phase

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THEME: Detecting AI-Generated Fraud Content

PROBLEM STATEMENT

- Everyday, millions of people fall victim to online threats they never saw coming. The threats such as a convincing email from a bank asking for profile modification / password change or SMS with link.

- What has changed and who is behind these attacks:

  Generative AI has handed cybercriminals a weapon they have never had before — the ability to craft phishing emails that read like they were written by a colleague, spin up spoofed websites in minutes, clone voices for deep fake scams, and embed prompt injection payloads that silently hijack AI systems.

  Traditional rule-based filters were never built for this. They flag yesterday's threats, miss today's, and bury analysts in false alarms that erode trust in the entire detection pipeline.

- We saw this gap and decided to build something different:

  A software to detect and prevent autonomous AI attacks (online frauds) using human-augmented AI defense.

SOLUTION

How it solves the problem

- 2-layer AI pipeline detects phishing, credential leaks & malicious URLs.
- Replaces static blacklists with context-aware LLM reasoning.
- Catches AI-generated fraud and hidden intent invisible to rule-based filters.

Impact Metrics

- Precision / Recall / F1 on phishing detection.
- False positive rate.
- Email latency (ML vs LLM path).
- Escalation rate (human review %).
- Compute savings from ML pre-filtering.

Tech Stack

- Python
- sklearn
- TF-IDF
- Ollama
- Regex
- Pandas
- Joblib

Assumptions & Decision Points

- Phi3 chosen - open-source, zero cost, local execution.
- TF-IDF over deep embeddings - lightweight & interpretable.
- Regex for credentials - no model overhead needed.
- Scope limited to email input.

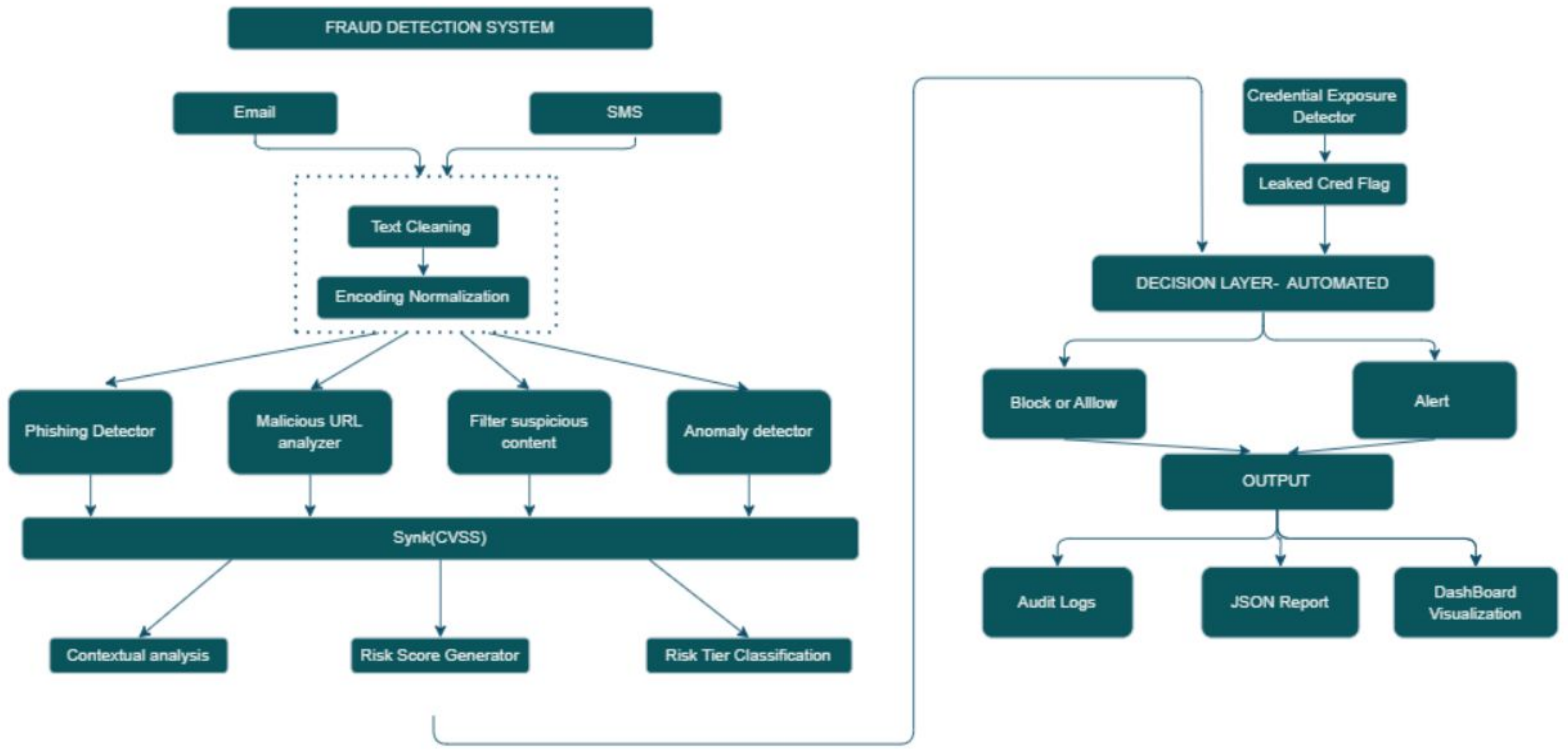
Implementation

- Single machine setup.
- Modular layers - each independently testable.
- Every verdict includes a plain-English explanation for audit trails.

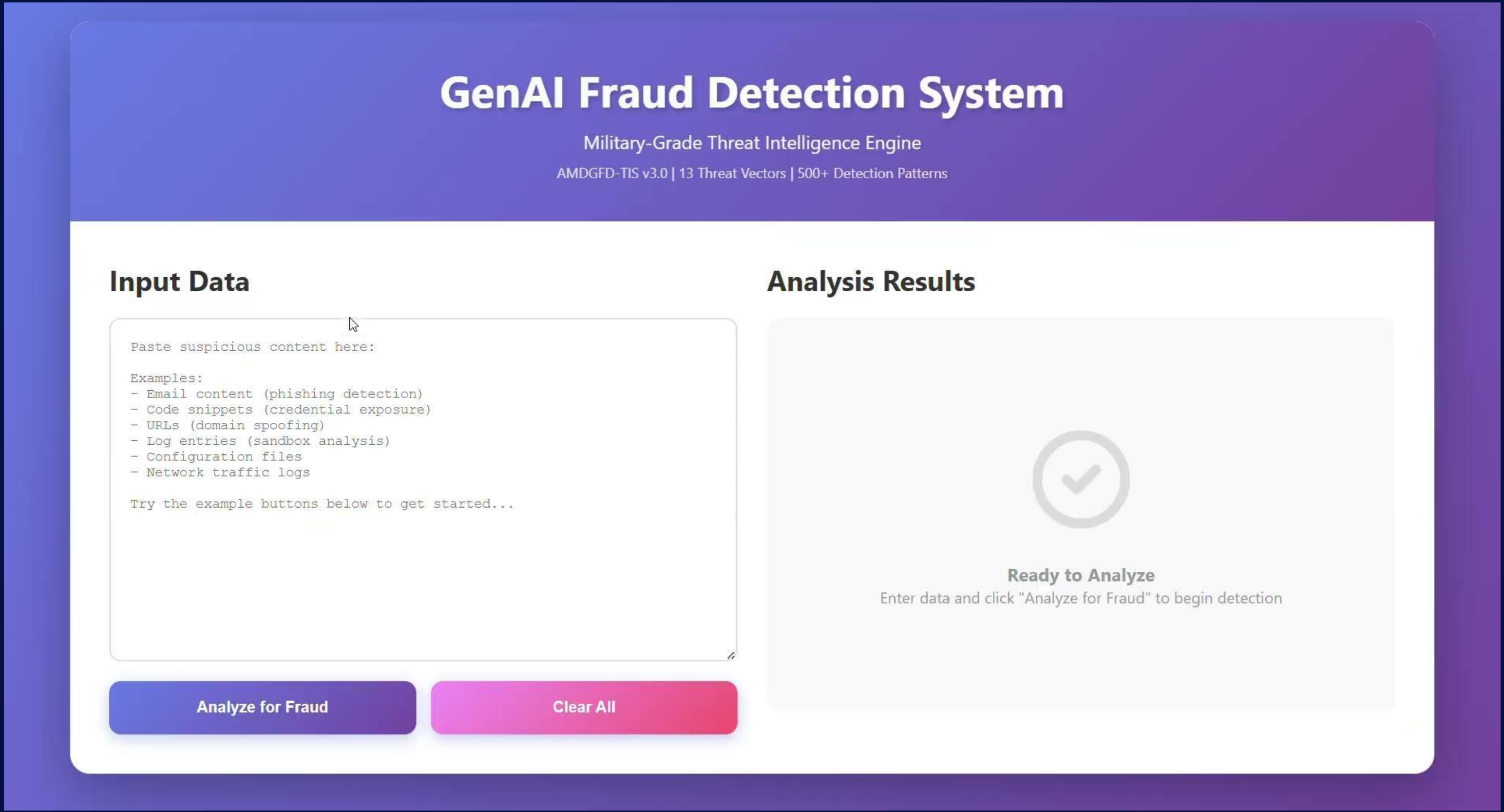
Scalability & Usability

- LLM layer horizontally scalable.[4]
- REST API ready for bank email system integration.[5]
- JSON output connects directly to SIEM dashboards.

METHODOLOGY



WORKING PROTOTYPE



ATTACHMENTS

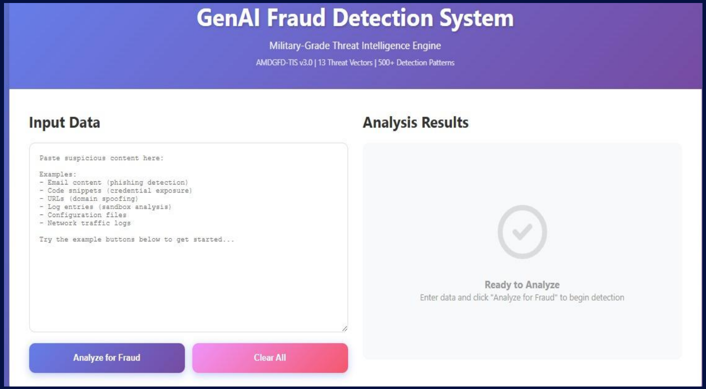


Fig 1: User interface

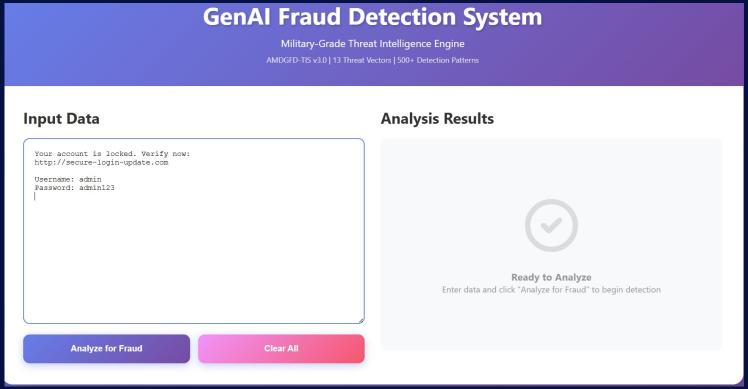


Fig 2: User Input

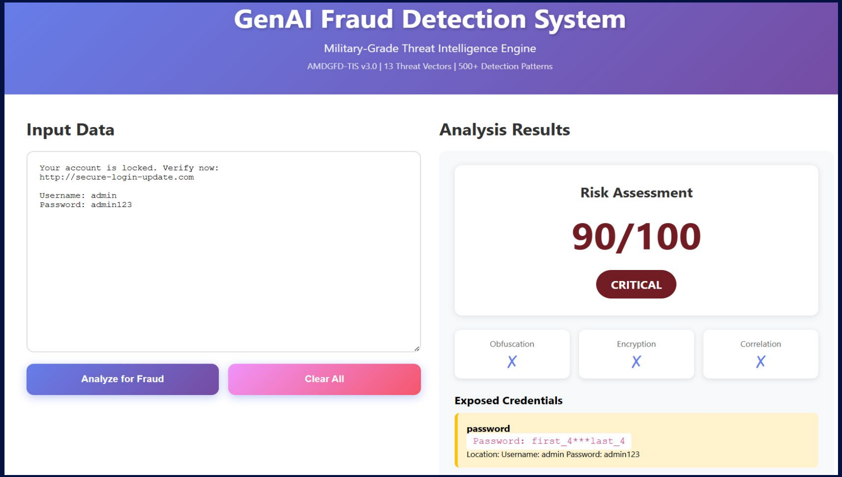


Fig 3: Result Analysis

Impact, Future Scope & Business Relevance

Business Relevance

- Deployable in banks, fintech companies, and enterprises.
- Supports privacy-first architecture with online deployment.
- Scalable for integration with Security Operations Centers (SOC) and email gateway systems.

Future Scope

- Image-based fraud detection and Deepfake voice detection.
- Behavioral analytics & SIEM integration.
- Continuous retraining against evolving GenAI threats.
- Prompt injection detector.
- For production at Barclays scale, swap Phi-3 for a cloud API like GPT-4 or Claude which return sub-second.

Impact

- Reduces AI-driven phishing and credential theft.
- Minimizes financial fraud and false positives.
- Strengthens trust in digital systems.

Novelty

- Explainable risk scoring pipeline.
- **Hybrid ML + LLM architecture** — Combines fast TF-IDF pre-screening with deep contextual LLM analysis, achieving 98% compute savings vs full LLM scan.
- **Explainable AI verdicts.**

References

1. APWG Phishing Activity Trends Report 2025
2. IBM Cost of a Data Breach Report 2025
3. RBI Financial Fraud Data FY 2024-25
4. Kubernetes Horizontal Pod Autoscaler (HPA)
5. OpenAPI 3.1 Specification